Topic 2.8 - Inverse Functions

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: ____________ Class: ____________ Date: ____________

Why this topic matters

An inverse function reverses a one-to-one relationship. The domain of the original becomes the range of the inverse, and the range of the original becomes the inverse domain. Restrictions are part of the definition, not optional details.

Learning targets

- Determine whether a function is invertible on a stated domain.
- Construct and verify inverse functions algebraically, numerically, and graphically.
- Interpret inverse inputs, outputs, domains, ranges, and restrictions in context.

Language and structure

Term	Meaning in this topic
one-to-one	A function in which no two allowed inputs produce the same output.
inverse function	A function that reverses the input-output pairing of another function.
horizontal line test	A graphical test for whether a function is one-to-one.
domain restriction	A smaller domain chosen so a function becomes one-to-one.
inverse verification	Showing $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$ on the appropriate domains.

Launch: make a claim before calculating

Launch investigation

Reasoning first

A machine maps an input setting to an output reading. Settings 2 and 8 both produce reading 11. Can the machine's rule have an inverse function on a domain containing both settings? Explain the practical meaning.

Concept 1: Invertibility and restrictions

Core idea

A function has an inverse function exactly when it is one-to-one on its stated domain.

- Use the horizontal line test on a graph.
- A quadratic is not one-to-one on all real numbers but can be restricted to one side of its vertex.
- State the restricted domain before writing the inverse.

Worked example - annotate the reasoning

Problem. Determine whether $f(x) = (x-2)^2 + 1$ is invertible on $\mathbb{R}$ and on $[2, \infty)$.
Model reasoning. On all real numbers it is not one-to-one because symmetric inputs give the same output. On $[2, \infty)$ it is increasing and one-to-one, so an inverse exists.

Check your understanding

Independent

Give a domain restriction that makes $g(x) = -(x+1)^2 + 5$ invertible.

Concept 2: Construct and verify an inverse

Core idea

Write $y = f(x)$, interchange x and y, solve for y, and carry the domains and ranges with the result.

- Linear and rational functions often invert algebraically without a branch choice.
- Even roots introduce a plus-or-minus choice that is resolved by the original domain.
- Verify by composition and by checking that points (a, b) become (b, a).

Worked example - annotate the reasoning

Problem. Find the inverse of $f(x) = \dfrac{2x+5}{x-3}$.
Model reasoning. Let $y = (2x+5)/(x-3)$. Then $yx - 3y = 2x + 5$, so $x(y-2) = 3y + 5$ and $x = (3y+5)/(y-2)$. Thus $f^{-1}(x) = \dfrac{3x+5}{x-2}$, with inverse domain $x \neq 2$.

Check your understanding

Independent

Find the inverse of $h(x) = (x-2)^2 + 1$ on $x \geq 2$.

Concept 3: Graph and context

Core idea

The graphs of f and f^{-1} are reflections across the line $y = x$.

- x- and y-intercepts swap roles when they are defined.
- In context, the inverse answers the reverse question: output-to-input rather than input-to-output.
- Units swap: if f maps hours to distance, f^{-1} maps distance to hours.

Worked example - annotate the reasoning

Problem. A conversion rule maps temperature C in degrees Celsius to $F(C) = 1.8C + 32$ degrees Fahrenheit. Find and interpret the inverse.
Model reasoning. $F^{-1}(x) = (x-32)/1.8$. It takes a Fahrenheit temperature as input and returns the corresponding Celsius temperature.

Check your understanding

Independent

If $(4, -7)$ lies on f, what point lies on f^{-1}?

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

Find the inverse of $f(x) = 5x - 9$.

Exit ticket 2

No calculator

State the inverse domain of $g(x) = \sqrt{x+4} - 2$.

Exit ticket 3

No calculator

What graph transformation connects f and f^{-1}?